

youkined from <https://github.com/Nicicalu/OST-CN2-Spik>

| OSPF (IGP)                                                                                                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Equal Cost MultiPath (ECMP) load balancing, fast convergence, widely used. Uses IP Port 89 on Layer 3. Uses 224.0.0.5 (all routers), 224.0.0.6 (all DR,BDR)            |
| <b>DR Election:</b> Highest interface priority → if tie, highest Router ID. Priority 0 = ineligible for DR/BDR. No preemption. BDR = same rules, second best candidate |
| ROUTER OPERATION                                                                                                                                                       |
| – Establish neighbor adjacencies and exchange LSAs                                                                                                                     |
| – Build the Link-State Database (LSDB)                                                                                                                                 |
| – Run Dijkstra's SPF algorithm:                                                                                                                                        |
| – <b>Inter-area change:</b> SPF recalculation                                                                                                                          |
| – <b>Inter-area change:</b> No SPF needed → ABR handles updates                                                                                                        |
| – Build the routing table from SPF results                                                                                                                             |
| AREAS                                                                                                                                                                  |
| – AS split into <b>areas</b> (sub-domains), each with a 32-bit Area ID (e.g., 0.0.0.0 = Area 0)                                                                        |

**Backbone Area (Area 0):** Core of the OSPF domain (must exist). Must connect to all other areas (directly or via virtual links). Must be contiguous (no disjointed segments). Should not contain end-user networks

**Non-Backbone Areas:** Connect end users and local resources. All inter-area traffic must transit the backbone

| ROUTER TYPES                                                                                                                                                 |
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| <b>Area Border Router (ABR):</b> Connects two or more OSPF areas; must have interfaces in backbone, 1 OSPF DB per Area                                       |
| <b>Internal Router:</b> All interfaces belong to the same area (non-backbone)                                                                                |
| <b>Backbone Router:</b> At least one interface in Area 0, Includes ABRs and routers internal to the backbone                                                 |
| <b>AS Boundary Router (ASBR):</b> Connects to external AS/Network (e.g., BGP). Advertises external routes into OSPF. Can be in backbone or non-backbone area |

| DESIGN RULES                                                                |
|-----------------------------------------------------------------------------|
| <b>Rule 1:</b> Backbone (Area 0) must be contiguous → no partitions allowed |
| <b>Rule 2:</b> Every non-backbone area must connect to Area 0               |

| LSA TYPE OVERVIEW                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Type 1 – Router-LSA:</b> Sent by all routers, lists directly connected links (so all outgoing interfaces with state and cost) (intra-area only)   |
| <b>Type 2 – Network-LSA:</b> Sent by DR, lists all routers and DR in broadcast/multi-access network (intra-area only)                                |
| <b>Type 3 – Summary-LSA:</b> Sent by ABRs, advertises networks from other areas, flooded in all the areas that are not "totally stubby" (inter-area) |
| <b>Type 4 – ASBR-Summary:</b> Sent by ABRs, advertises path to ASBRs (inter-area)                                                                    |
| <b>Type 5 – AS-External-LSA:</b> Advertises external routes (e.g., BGP); flooded to all non-sub areas – only normal areas                            |
| <b>Type 7 – NSSA External:</b> Like Type 5 but used inside NSSA; converted to Type 5 by ABR                                                          |

| AREA TYPES (ALLOWED LSA TYPES)                    |                                                               |
|---------------------------------------------------|---------------------------------------------------------------|
| <b>Standard (1,2,3,4,5):</b> Normal               | <b>Not-So-Stubby Area – NSSA (1,2,3,7):</b>                   |
| <b>Stub Area (1,2,3):</b>                         | <b>(1,2,3,7):</b>                                             |
| – Blocks external LSAs (Type 5)                   | – Like stub area, but allows one ASBR inside                  |
| – ABR injects a default route (0.0.0.0)           | – External routes use LSA Type 7 (converted to Type 5 by ABR) |
| – Supports LSA Types 1–3                          |                                                               |
| <b>Totally Stubby Area (1,2):</b>                 | <b>Totally NSSA (1,2,7):</b>                                  |
| – Blocks external (Type 5) and summary (Type 3/4) | – Like NSSA but blocks Type 3 so gets default route from ABR  |
| – Only allows default route from ABR              |                                                               |

| PACKET TYPES                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Type 1 – Hello:</b> Used to discover, maintain, and verify neighbors; forms adjacencies. Also for election of DR,BDR in broadcast networks. Contains network mask(of sending routers' interface), Hello interval(p2p,broadcast: default=10s), Options, Priority(for election), Router dead interval(default=40s), DR/BDR IP, Neighbors |
| <b>Type 2 – Database Description (DBD/DD):</b> Exchange summaries of LSAs during adjacency formation (headers only). Contains Interface Max. MTU, Options, I/M/MS bits (Initial, More, Master-slave bit), DD Sequence num, LSA Header                                                                                                     |
| <b>Type 3 – Link State Request (LSR):</b> Sent when a router needs specific LSAs listed in the DBD. Contains Link State Type (router/network), Link State ID, Advertising Router (sender address)                                                                                                                                         |
| <b>Type 4 – Link State Update (LSU):</b> Used to flood new or updated LSAs. Contains Number of LSAs, full LSAs information                                                                                                                                                                                                                |
| <b>Type 5 – Link State Acknowledgment (LSAck):</b> Confirms receipt of LSAs to ensure reliable flooding. For this you send LSAck or implicitly by sending LSU with same info back. Many acks may be grouped together to a single LSAck.                                                                                                   |

| SUB-PROTOCOLS                                                                                                                                                    |
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| <b>Hello Protocol:</b>                                                                                                                                           |
| – Used for neighbor discovery and parameter negotiation.                                                                                                         |
| – Maintains logical adjacencies on P2P, P2MP, and virtual links.                                                                                                 |
| – Elects DR/BDR on broadcast and NBMA networks.                                                                                                                  |
| – Continuously sends hello packets to maintain bidirectional connectivity; failure to receive = neighbor down (in agreed router dead interval at initialization) |
| <b>Database Sync Protocol:</b>                                                                                                                                   |
| – Syncs LSDB using Database Description (DBD) packets with only LSA headers.                                                                                     |
| – Uses I-bit (initial), M-bit (more), and MS-bit (master/slave).                                                                                                 |
| – <b>ExStart:</b> Bi-direction; highest Router-ID = master. Determine initial seq nr                                                                             |
| – <b>Exchange:</b> Exchange of DBD packets (LSA headers).                                                                                                        |
| – <b>Loading:</b> Missing LSAs are requested.                                                                                                                    |
| – <b>Full:</b> Databases fully synchronized.                                                                                                                     |

| OSPF ROUTING AND ECMP                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| – Each router runs Dijkstra per area; link cost = metric from LSAs (1–65535)                                                                              |
| – OSPF prefers more specific match (CIDR), and if then still multiple: intra-area > inter-area > external                                                 |
| – Routes added to RIB/FIB based on computed next hops                                                                                                     |
| – <b>ECMP:</b> Modified Dijkstra supports Equal-Cost MultiPath if multiple paths have same cost → routes added with multiple next-hops for load balancing |

| OSPF ROUTE SELECTION                                                                                          |
|---------------------------------------------------------------------------------------------------------------|
| <b>Intra-Area (O):</b> Source and dest in same area; routes from Type 1 and 2 LSAs                            |
| <b>Inter-Area (O IA):</b> Source and dest in different areas within same AS; via Type 3 LSAs through backbone |

| <b>External (E1/E2):</b> Dest outside AS; info injected by ASBR via redistribution                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| – <b>E1:</b> Total = external + internal OSPF cost                                                                                                  |
| – <b>E2:</b> Only external cost (default)                                                                                                           |
| <b>Preference order:</b> More specific route > Intra-area > Inter-area > E1 > E2                                                                    |
| <b>Cost calculation:</b> Cost = Reference Bandwidth/default 100 Mbps/Interface Bandwidth                                                            |
| IS-IS (IGP)                                                                                                                                         |
| <b>CLNS – CONNECTIONLESS NETWORK SERVICES</b>                                                                                                       |
| <b>CLNP:</b> ISO Layer 3 datagram service; supports CLNP ES-IS, IS-IS. (EtherType 0xFFFE)                                                           |
| <b>IS-IS:</b> Link-state routing protocol (Layer 3); forms adjacencies with ES-IS; designed for CLNP but extended (Integrated IS-IS) to support IP. |
| <b>Integrated IS-IS:</b> Allows IS routing with IS-IS; used widely by service providers even without CLNP.                                          |

| NET ADDRESSING (TYPE OF NSAP ADDRESS)                                  |
|------------------------------------------------------------------------|
| <b>NSAP:</b> Network Service Access Point                              |
| <b>NET = Network Entity Title:</b> Unique router identifier in IS-IS   |
| <b>Format:</b> 49.AAAA.BBBB.BBBB.BBBB.00                               |
| 49.AAAA... Area ID (variable length)                                   |
| BBBB.BBBB.BBBB: System ID (usually 6 bytes = unique router ID)         |
| 00: N-selector (NSEL) (always 00 for routers)                          |
| <b>System ID:</b> Must be unique per router (often based on lo-IP/MAC) |
| Example: 49.0001.1921.6800.1024.00 based on IP 192.168.1.24            |
| <b>Area ID:</b> Used for routing hierarchy (like OSPF areas)           |

| IS-IS PACKET TYPES (ALL IN L1 OR L2)                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IIH (Hello):</b> Builds and maintains adjacencies; includes system ID, holding time, prio                                                |
| – Built from 3 functions: <b>discover, build, maintain</b>                                                                                  |
| – <b>Interval:</b> 10s (default); DIS sends every 3.3s on LANs                                                                              |
| – <b>Multipoint:</b> Missed Hello limit → Holdtime + interval × Multiplier(default=3)                                                       |
| <b>Multicast:</b>                                                                                                                           |
| – 01-80-c2-00-00-14 (All L1ISs)                                                                                                             |
| – 01-80-c2-00-00-15 (All L2ISs)                                                                                                             |
| <b>LSP (Link State PDU):</b> Contains topology info including prefixes with costs; flooded throughout the area → similar to OSPF LSA Type 1 |
| <b>CSNP (Complete SNP):</b> Sent by DIS; lists all known LSPPs (used for database sync)                                                     |

| <b>PSNP (Partial SNP):</b> Used to request missing LSPs or acknowledge received LSPs |
|--------------------------------------------------------------------------------------|
| <b>IS-IS Packet Structure:</b> Common header + TLVs                                  |
| ROUTER TYPES                                                                         |
| <b>L1 Router:</b> Only within one area; no inter-area routing                        |
| <b>L2 Router:</b> Backbone router; routes between areas                              |
| <b>L1/L2 Router:</b> Acts as both; separates databases, redistributes between levels |

| BROADCAST/MULTI-ACCESS LINKS: DIS – DESIGNATED IS                            |
|------------------------------------------------------------------------------|
| – Required on broadcast links (no DIS on p2p)                                |
| Sends periodic <b>CSNPs</b> to ensure DB sync, creates <b>pseudonode LSP</b> |
| No backup DIS in IS-IS                                                       |
| DIS ELECTION                                                                 |
| – 1) Highest interface priority (0–127) Cisco default = 64                   |
| – 2) Highest SNPA (MAC-Address)                                              |

| <b>Preamble:</b> Enabled – higher prio router automatically takes ver the DIS Role |
|------------------------------------------------------------------------------------|
| POINT-TO-POINT LINKS (NO DIS)                                                      |
| <b>CSNP:</b> Sent once at adjacency startup                                        |
| <b>LSP:</b> Advertises topology changes (link-state info)                          |
| <b>PSNP:</b> Acknowledges received LSPs or requests missing ones                   |

| PATH SELECTION                                                                                |
|-----------------------------------------------------------------------------------------------|
| <b>Path Selection Order (in IS-IS) – Lower Metric better:</b>                                 |
| – L1 intra-area routes                                                                        |
| – L2 intra-area routes                                                                        |
| – Leaked L2 → L1 (internal metric)                                                            |
| – L1 external (external metric)                                                               |
| – L2 external (external metric)                                                               |
| – Leaked L2 → L1 (external metric)                                                            |
| LEVEL 1 ROUTING                                                                               |
| – Intra-area routing only (like OSPF intra-area)                                              |
| – L1 routers use the closest L1/L2 router for inter-area traffic                              |
| <b>L1/L2 routers:</b>                                                                         |
| – Do not advertise L2 routes into the L1 area (unless route leaking is active)                |
| – Set <b>Attached bit</b> to signal L2 connectivity to backbone                               |
| – L1 routers install a <b>default route</b> to nearest L1/L2                                  |
| – L1 area like OSPF Totally Stubby Area                                                       |
| – <b>Distribution Bit:</b> Set to 'up' (1) on L2 → L1 leaks; blocks re-advertisement L1 → L2. |
| – <b>Route-Leaking</b> injects a more specific route into L1 to improve routing               |

| LEVEL 2 ROUTING                                                        |                      |                   |
|------------------------------------------------------------------------|----------------------|-------------------|
| Routing between areas (inter-area)                                     |                      |                   |
| L1/L2 routers inject L1 routes into L2 topology                        |                      |                   |
| L1 routes are redistributed into L2 with L1 metric preserved in L2 LSP |                      |                   |
| IS-IS VS OSPF                                                          |                      |                   |
| Feature                                                                | IS-IS                | OSPF              |
| Layer                                                                  | L2 (CLNS)            | L3 (IP, proto 89) |
| Encapsulation                                                          | No IP, uses TLVs     | IP packets        |
| Hello Type                                                             | IIH                  | Hello packet      |
| Area Model                                                             | L1/L2                | Backbone + Areas  |
| Metric                                                                 | Cost (default 10)    | Cost (bandwidth)  |
| Router ID                                                              | System ID (6B)       | 32-bit Router ID  |
| Adj. Types                                                             | L1, L2, L1/L2        | DR/BDR, P2P       |
| LSDB                                                                   | Per level (L1/L2)    | Per area          |
| Scaling                                                                | Large-scale ISP core | Enterprise/campus |
| Routing Info                                                           | TLVs (flexible)      | Fixed LSA types   |

| AUTONOMOUS SYSTEM NUMBERS (ASN)                                 |
|-----------------------------------------------------------------|
| – Unique ID for each AS; required for Internet routing with BGP |

| – Private ranges:                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------|
| – 64512–65535 (legacy 16-bit)                                                                                                       |
| – 4200000000–4294967294 (32-bit)                                                                                                    |
| BGP PEERING / NEIGHBORS                                                                                                             |
| – Two routers with a BGP TCP session (port 179) are called peers or neighbors                                                       |
| – Each BGP router is a <b>BGP speaker</b>                                                                                           |
| – BGP exchanges routing info between ASes (loop-free, policy-based)                                                                 |
| – Supports CIDR, route aggregation; decisions based on policies/rules                                                               |
| PATH ATTRIBUTES                                                                                                                     |
| Used for route control and policy enforcement                                                                                       |
| <b>Well-known mandatory:</b> Always present (e.g., AS-Path, Origin, Next Hop)                                                       |
| <b>Well-known discretionary:</b> Optional but recognized by all (e.g., Local Pref, Atomic Aggregate)                                |
| <b>Optional transitive:</b> Passed between ASes (e.g., Community, Aggregator)                                                       |
| <b>Optional non-transitive:</b> Not passed across ASes (e.g., MED, Weight, Originator ID, Cluster ID/list)                          |
| <b>NLRI:</b> Routing table info: prefix, prefix length, and associated path attributes                                              |
| LOOP PREVENTION                                                                                                                     |
| BGP uses <b>AS-Path</b> (list of ASNs) to detect loops. If a router sees its own ASN in a received route, it discards it.           |
| <b>AS-Override:</b> Allows reuse of same ASN across different customer sites (e.g., Swisscom); rewrites ASN to avoid loop detection |

| BGP MESSAGES                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------|
| <b>OPEN:</b> Establishes session; includes version, ASN, Hold Time, BGP Identifier, optional params                               |
| – <b>Hold Time:</b> Heartbeat in seconds (default 180s, Cisco), reset by KEEPALIVE/UPDATE; 0 = session down                       |
| – <b>BGP Identifier:</b> 32-bit Router-ID, manually set or highest loopback/active IP; used for loop prevention                   |
| <b>KEEPALIVE:</b> Sent every 1/3 of Hold Time (default 180s); ensures neighbor liveness (BGP doesn't rely on TCP keepalive)       |
| <b>UPDATE:</b> Advertises new routes, withdraws old ones, or both; includes NLRI (prefix + path attributes); can act as KEEPALIVE |
| <b>NOTIFICATION:</b> Sent on session error (e.g. hold timer expired); terminates session immediately                              |

| BGP NETWORK STATEMENTS                                                             |
|------------------------------------------------------------------------------------|
| – Purpose: Advertise specific prefixes to BGP peers (does not activate interfaces) |
| – Prefix must exist exactly in the RIB (from static, connected, or learned route)  |
| – Attributes (e.g., origin, next-hop, MED) depend on how the route exists in RIB   |
| – BGP advertises only the best path for a prefix to peers, even if multiple exist  |
| BEST PATH CALCULATION                                                              |
| – BGP maintains all received paths per prefix but advertises only the best one     |
| – Best path is installed in RIB; recalculated on:                                  |
| – Next-hop reachability change                                                     |
| – Interface failure to eBGP peer                                                   |
| – Redistribution change                                                            |
| – New/withdrawn path received                                                      |
| – <b>Influence:</b>                                                                |
| – Outbound BGP policy → inbound traffic behavior                                   |
| – Inbound BGP policy → outbound traffic behavior                                   |

|                                                                                     |
|-------------------------------------------------------------------------------------|
| <b>BGP BEST PATH SELECTION (IN ORDER):</b>                                          |
| 1) Prefer highest <b>Weight</b> (Cisco-specific, local to router)                   |
| 2) Prefer highest <b>Local Preference</b> (global within AS)                        |
| 3) Prefer routes <b>originated by the router</b> (only small i in path, NH 0.0.0.0) |
| 4) Prefer <b>shorter AS path</b> (only length is compared)                          |
| 5) Prefer <b>lowest origin type:</b> IGP < EGP < Incomplete (I on origin)           |
| 6) Prefer <b>lowest MED</b> (Multi-Ext Discriminator) (also called metric)          |
| 7) Prefer <b>external (eBGP)</b> over internal (iBGP)                               |
| 8) For iBGP: prefer path with <b>lowest IGP metric</b> to next-hop                  |
| 9) For eBGP: Prefer <b>oldest</b> (more stable) path                                |
| 10) Prefer <b>lowest BGP router ID</b>                                              |
| 11) Prefer path from <b>lowest neighbor IP address</b>                              |

| ROUTE FILTERING                                                                                        |
|--------------------------------------------------------------------------------------------------------|
| – Filters control which routes are received/advertised                                                 |
| – Used for security, traffic shaping, memory optimization                                              |
| – Tools: <b>prefix-list</b> (IP), <b>filter-list</b> (AS-path), <b>route-map</b> (flexible match/set)  |
| BGP COMMUNITIES                                                                                        |
| – 32-bit optional, transitive tag (e.g. <b>ASN-value, 65000:100</b> )                                  |
| – Used to mark routes for policy control across ASes                                                   |
| – Can be added, modified, or removed at each hop                                                       |
| iBGP SCALABILITY                                                                                       |
| – iBGP does not re-advertise routes between iBGP peers → full mesh required (cause no loop prevention) |
| – Session count = $n \times (n-1) / 2$ (e.g., 5 routers = 10 sessions, 10 = 45)                        |

| ROUTE REFLECTORS (RR)                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------|
| – Solves iBGP full-mesh scaling by allowing selective route reflection                                                        |
| – Clients only peer with RR; unaware they're clients                                                                          |
| – <b>RR Rules:</b>                                                                                                            |
| – From <b>non-client</b> → advertise to <b>clients only</b>                                                                   |
| – From <b>client</b> → advertise to all (clients & non-clients)                                                               |
| – From <b>eBGP peer</b> → advertise to all (clients & non-clients)                                                            |
| – Only the RR needs specific confg - clients remain unaware of route reflection. This eliminates the need for full iBGP mesh. |

| PEERING VS. TRANSIT                                                                       |
|-------------------------------------------------------------------------------------------|
| <b>Transit:</b> ISP provides full reachability (paid relationship)                        |
| <b>Peering:</b> ISPs exchange selected routes; equal relationship, usually unpaid         |
| <b>AS Path Filtering:</b> to avoid getting transit: ip as-path access-list 10 permit ^ \$ |
| INTERNET EXCHANGE POINT (IXP)                                                             |
| – Facility where networks exchange traffic via BGP peering                                |
| – Reduces transit costs, latency, and offloads upstream links                             |
| PUBLIC PEERING                                                                            |
| – Members peer via shared switch fabric and a <b>route server</b>                         |
| – Route server distributes routes but stays out of data path (NEXT_HOP unchanged)         |
| – Minimal policy control; one BGP session to route server                                 |
| – Simplified setup (one legal contract)                                                   |
| PRIVATE PEERING                                                                           |
| – Direct BGP sessions between two parties (1:1)                                           |
| – May use public or private interconnects                                                 |
| – Full policy control per neighbor                                                        |
| – Requires one session and legal contract per peer                                        |

| <b>Examples:</b> Equinix, SwissIX (non-profit)                                   |
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| ENTERPRISE CONNECTIVITY OPTIONS                                                  |
| <b>Single-Homed:</b> One ISP, one link (BGP or static); simple but no redundancy |

| <b>Dual-Homed:</b> One ISP, two links (or routers); redundancy within same provider                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Multihomed:</b> Multiple ISPs; improved redundancy and routing control, but avoid being a transit → advertise only customer-owned prefixes                            |
| <b>Dual-Multihomed:</b> Multihomed, but two links per ISP                                                                                                                |
| TRAFFIC ENGINEERING (TE)                                                                                                                                                 |
| <b>Outbound TE (Local Pref):</b> Set higher local pref to prefer exit path; affects outbound traffic; highest wins                                                       |
| <b>Inbound TE (MED):</b> Signal entry preference with MED; lowest wins; only works if peer honors it                                                                     |
| <b>Inbound TE (AS-Path Prepending):</b> Add own ASN multiple times on backup path; shortest AS-path wins                                                                 |
| <b>TE Limitation:</b> AS controls outbound (e.g. local pref); inbound control limited                                                                                    |
| ISPs may ignore MED                                                                                                                                                      |
| <b>TE with Aggregate:</b> Prefer primary ISP with summarized routes; advertise specific prefixes on backup for fallover                                                  |
| <b>Aggregate Impact:</b> Longest-match wins → specific prefixes may steer traffic to alternate ISP; avoid provider-owned aggregates                                      |
| RPKI – RESOURCE PUBLIC KEY INFRASTRUCTURE                                                                                                                                |
| – Framework to validate that a prefix is legitimately originated by a specific ASN                                                                                       |
| – Prevents route hijacking and accidental mis-originations (route leaks)                                                                                                 |
| – RPKI validates origin → not the 5 RRs                                                                                                                                  |
| KEY COMPONENTS                                                                                                                                                           |
| <b>Trust Anchors (TA):</b> Root CAs of the AS-RPs; issue certs for resource holders                                                                                      |
| <b>ROA – Route Origin Authorization:</b> Digitally signed object that authorizes ASN to announce a prefix (contains AS, prefix that AS can originate, max prefix length) |
| <b>RPKI Validators:</b> Software that downloads, verifies, and stores Validated ROA Validations (VRPs)                                                                   |

| RPKI VALIDATION STATES                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Valid:</b> Prefix + ASN match ROA                                                                                                                   |
| <b>Invalid:</b> Prefix found, but ASN mismatch or prefix too long                                                                                      |
| <b>Not Found / Unknown:</b> No matching ROA                                                                                                            |
| VALIDATOR DETAILS                                                                                                                                      |
| – Use TALS (Trust Anchor Locators) to fetch data from RIRs                                                                                             |
| – Validate cryptographic signatures (via X.509 certs with RFC 3779)                                                                                    |
| – Outputs VRPs; invalid objects are discarded                                                                                                          |
| – Update frequency: ~24h (recommended), ~30–60min (practice)                                                                                           |
| – RRDP (RFC 8182) replacing rsync (uses HTTPS)                                                                                                         |
| BGP MONITORING                                                                                                                                         |
| – Event tracking, BGP hijack detection, Route leak detection, RPKI status check, Reachability tracking, AS path change tracking, AS path visualization |

| NETWORK DESIGN                                                                                  |
|-------------------------------------------------------------------------------------------------|
| <b>Pillars:</b> Scalability, Speed, Availability, Security, Manageability → overall Cost        |
| AVAILABILITY CONCEPTS (MOST IMPORTANT REQUIREMENT)                                              |
| <b>MTBF:</b> Mean Time Between Failures, <b>MTTR:</b> Mean Time to Repair                       |
| – $A = \frac{MTBF}{MTBF + MTTR}$ MTBF combined: $(\sum_{n=1}^N \frac{1}{MTBF_n})^{-1}$ parallel |
| – Lower <b>MTTR</b> + higher MTBF = more availability                                           |
| REUNDANCY                                                                                       |
| – Adds reliability, decreases MTBF but increases MTTR and complexity                            |
| – Balance: resilience vs. manageability                                                         |

**Backup Paths:** Duplicate devices/links on primary path, build extra links for redundancy, consider backup link capacity, consider failover speed

**Load Balancing:** ECMP, EtherChannel, Port-Channel

| HIERARCHICAL DESIGN                                                                     |
|-----------------------------------------------------------------------------------------|
| <b>Access:</b> Connect end devices, high port count, port security, L2, QoS marking     |
| <b>Distribution:</b> L3, policy control, HSRP/VRRP, loop protection, small fault domain |
| <b>Core:</b> High-speed backbone, L3 only, no policy, scalable/redundant, no security   |
| <b>Collapsed Core:</b> Combines Core + Distribution (small/medium networks)             |
| FLAT DESIGN                                                                             |
| – Any-to-any; small networks, MPLS/LAN setups                                           |
| – Lacks scalability and control                                                         |
| FABRIC DESIGN                                                                           |
| – Modern design using <b>Underlay/Overlay</b>                                           |
| – Underlay: transport (e.g., IP, MPLS)                                                  |
| – Overlay: logical virtual topology (e.g., EVPN)                                        |
| ENTERPRISE CANPUS                                                                       |
| – 100s/1000s of users, multiple buildings, one physical location                        |
| – Multiple interconnected LANs, connected via Ethernet and Wireless                     |
| FHRP – FIRST HOP REDUNDANCY PROTOCOLS                                                   |
| – Ensures default gateway is always reachable, PCs can only have one                    |
| – Enables fast failover during router failure                                           |

|                                                                                                                                                                                                                      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>VRPP – Virtual Router Redundancy Protocol (Multivendor):</b> Shared virtual IP, real MACs per router, Master router handles forwarding                                                                            |
| <b>HSRP – Hot Standby Router Protocol (Cisco):</b> Shared virtual IP + virtual MAC, One active, others in standby → faster                                                                                           |
| <b>GLBP – Gateway Load Balancing Protocol (Cisco):</b> Load balancing + redundancy, Shared virtual IP + multiple virtual MACs, Roles: AVG: Answers ARP and sends virtual MAC addresses of AVFs AVF: Forwards traffic |

| DATA CENTER DESIGNS                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ToR - Top of Rack:</b> switches per rack, less cabling, easy expansions/exchanges per 'rack', scalable glass fiber, ideal for high service density (full racks); But more switches, more ports, more L2 Srv-2-Srv traffic, more STP to be managed |
| <b>EoR - End of Row:</b> 1 switch per row, less switches, higher utilization of ports, switches all at one place, better L2 availability between racks. But more cabling                                                                             |
| TRAFFIC DIRECTIONS                                                                                                                                                                                                                                   |
| <b>North-South:</b> Between external networks (client-server, in/out DC)                                                                                                                                                                             |
| <b>East-West:</b> Within DC (e.g., server-server, storage)                                                                                                                                                                                           |
| THREE-TIER DC                                                                                                                                                                                                                                        |
| – Access – Aggregation – Core (like Hierarchical)                                                                                                                                                                                                    |
| – Optimized for North-South traffic                                                                                                                                                                                                                  |
| – Not ideal for East-West communication                                                                                                                                                                                                              |
| LEAF-SPINE ARCHITECTURE                                                                                                                                                                                                                              |
| – Two-tier: Leaf switches (access) connect to Spines (core)                                                                                                                                                                                          |
| – High performance, low latency                                                                                                                                                                                                                      |
| – Scalable, ideal for East-West traffic                                                                                                                                                                                                              |

| MULTICAST                                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Use-cases 1-to-Many:</b> Streaming, software updates, music-on-hold                                                                         |
| <b>Use-cases Many-to-Many:</b> Gaming, VR, stock data, group chat                                                                              |
| <b>Benefits:</b> Efficient bandwidth, lower server/CPU load, no redundancy, supports multipoit apps                                            |
| <b>Properties:</b> UDP-based (no delivery guarantee, congestion control, or ordering) Apps must handle drops, duplicates, out-of-order packets |
| <b>Source:</b> Sends to group IP; doesn't need to join                                                                                         |
| <b>Receiver:</b> Must explicitly join group to receive traffic                                                                                 |
| MULTICAST ADDRESS RANGES                                                                                                                       |
| 224.0.0.0 – 224.0.0.255: Link-local, TTL = 1 (not forwarded by routers)                                                                        |
| 224.0.1.0 – 224.0.1.255: Reserved by IANA, routable                                                                                            |
| 232.0.0.0 – 232.255.255.255: Source-Specific Multicast (SSM)                                                                                   |
| 239.0.0.0 – 239.255.255.255: Administratively scoped (private multicast space)                                                                 |
| BROADCAST BASICS                                                                                                                               |
| L3 routes between subnets; L2 floods within same subnet                                                                                        |
| <b>L2 (Bridging):</b> MAC flit.filt.filt; switches flood to all ports in VLAN                                                                  |
| <b>L3 (Routing):</b>                                                                                                                           |
| – 255.255.255.255: local broadcast, not routed                                                                                                 |
| – Directed broadcast (e.g. 10.1.1.1.255): can be routed if enabled                                                                             |
| L2 MULTICAST: MAC MAPPING                                                                                                                      |
| 1) <b>Get IP:</b> Example multicast IP address: 239.5.5.5                                                                                      |
| 2) <b>Convert to Binary:</b> 239.5.5.5 = 11101111.00000101.00000101.00000101                                                                   |
| – Take only last 23 bits: 00000101.00000101.00000101                                                                                           |
| 3) <b>Map to MAC Prefix:</b> Use fixed MAC multicast prefix: 01:00:5E → Map last 23 bits to: 01:00:5E:05:05:05                                 |
| 4) <b>Final MAC Address:</b> 01:00:5E:05:05:05                                                                                                 |
| IGMP – INTERNET GROUP MANAGEMENT PROTOCOL (HOST TO FIRST-HOP-ROUTER)                                                                           |
| <b>Purpose:</b> Manages group membership for IPv4 multicast on each segment                                                                    |

|                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IGMPv1:</b>                                                                                                                                |
| – Basic join via query-response mechanism                                                                                                     |
| – No way for a host to leave a group explicitly                                                                                               |
| – Router sends general membership queries every 60s to 224.0.0.1                                                                              |
| – If no report is received, router removes group after timeout                                                                                |
| – Receiver has no knowledge of the multicast source                                                                                           |
| <b>IGMPv2:</b>                                                                                                                                |
| – Adds <b>Leave Group</b> message (faster pruning of unused traffic)                                                                          |
| – General queries to 224.0.0.1 every 125s 'Any hosts interested in any groups?'                                                               |
| – Supports <b>Group-Specific Queries</b> e.g. when someone leaves group ('anyone still interested in group xy?'), reducing broadcast overhead |
| – Still source-agnostic: receivers don't know who the source is                                                                               |
| <b>IGMPv3:</b>                                                                                                                                |
| – Adds <b>source filtering</b> (Include/Exclude lists)                                                                                        |
| – Enables Source-Specific Multicast (SSM) – receiver requests traffic only from selected source(s), no need for Rendezvous Point (RP) anymore |
| – Adds support for application-level access control and filtering                                                                             |
| – Can also be used in ASM (Any Source Multicast), but mainly with SSM                                                                         |

| IGMP SNOOPING                                                                                                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Without snooping: multicast = broadcast on VLAN                                                                                                                                |
| <b>With snooping:</b> switch listens to IGMP messages and builds a forwarding table                                                                                            |
| <b>Default:</b> snooping is enabled; switch needs IGMP Query to operate                                                                                                        |
| REVERSE PATH FORWARDING (RPF)                                                                                                                                                  |
| <b>RPF Check:</b> To avoid loops, verifies that a multicast packet arrives on the interface that a unicast packet destined for the multicast source would be forwarded out of. |
| RENDEZVOUS POINT (RP)                                                                                                                                                          |
| – Used in Shared Tree (*,G) setups with PIM-SM                                                                                                                                 |
| – RP acts as the common meeting point for sources and receivers                                                                                                                |
| – RPF check is performed toward the RP (not the source)                                                                                                                        |
| – Once the source is known, routers may switch to a Source Tree (S,G)                                                                                                          |
| SHARED TREE VS. SOURCE-BASED TREE                                                                                                                                              |
| <b>Shared Tree (*,G):</b>                                                                                                                                                      |
| – IGMP host sends a membership report (IGMP Join)                                                                                                                              |
| – Router adds (*,G) entry to multicast routing table                                                                                                                           |
| – * means 'any source' → source is unknown/unspecified                                                                                                                         |
| <b>Source-Based Tree (S,G):</b>                                                                                                                                                |
| – Built when router receives an (S,G) join/report from IGMP host                                                                                                               |
| – S = known multicast source; G = group                                                                                                                                        |
| – Router adds (S,G) to mroute table once source is known                                                                                                                       |

| PIM – PROTOCOL INDEPENDENT MULTICAST (ONLY BETWEEN ROUTERS)                                         |
|-----------------------------------------------------------------------------------------------------|
| – Relies entirely on the unicast routing table (RIB) for multicast forwarding decisions             |
| – Protocol-independent: works with static routes, OSPF, IS-IS, etc.                                 |
| PIM-DH – DENSE MODE (NO RP)                                                                         |
| <b>Push Model:</b> Floods multicast traffic to all interfaces; then prunes where no receivers exist |

| 1) <b>Flooding:</b> Source sends traffic → forwarded out all multicast-enabled links using unicast RIB                                                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2) <b>Distribution Tree:</b> Initially includes entire network (shared tree rooted at source)                                                              |
| 3) <b>Prune Messages:</b> Routers without interested receivers send prunes upstream to remove themselves from the tree                                     |
| 4) <b>State Maintenance:</b> Routers track source, receivers, interfaces to forward/prune per group                                                        |
| PIM-SM – SPARSE MODE (ASM, SSM)                                                                                                                            |
| <b>Push Model:</b> Multicast traffic is only sent where requested. Works with IGMP to detect interested receivers and uses unicast routing for forwarding. |
| 1) <b>Join/Prune:</b> Routers send explicit join/prune messages to request or stop receiving multicast for group (G) to other routers                      |
| 2) <b>Forwarding:</b> Routers only forward multicast packets for group (G) on interfaces from which explicit joins were received                           |
| ASM – ANY SOURCE MULTICAST (HOW IT WORKS WITH RP)                                                                                                          |
| – Works with <b>IGMPv1 or IGMPv2</b> → receiver does not know the source                                                                                   |
| – Receiver sends <b>IGMP Join (*,G)</b> to its first-hop router                                                                                            |
| – First-hop router forwards <b>PIM Join (*,G)</b> hop-by-hop toward the Rendezvous Point (RP)                                                              |
| – RP acts as a common meeting point for sources and receivers                                                                                              |
| – Sources send multicast traffic to the RP via a <b>PIM Register</b> tunnel                                                                                |
| – All routers in the multicast domain must know the RP location                                                                                            |

**Host Deletion:** When a host detaches, its ARP (default: 1500s) and MAC entry (default: 1800s) time out on the VTEP. Upon aging, the VTEP withdraws the host's MAC/L2VNI and IP/L3VNI advertisements.

- imp-null = networks are directly connected, no more label switching

**WAN**

Connects remote LANs via SPs for data/voice/video; key needs: bandwidth, control, design, resilience, mgmt. **Requirements:**

**Bandwidth:** App needs, peak usage, reserve for VoIP

**Bandwidth:** App needs, peak usage, reservation, over-  
**Control/Security:** Trust provider? No full control  
**Availability:** Redundancy, SLA for failures  
**Mgmt:** Inband vs out-of-band

- 64-bit RD + 32-bit IPv4 = 96-bit VPNv4 prefix
- transferring VPNv4 between PE router → Multiprotocol iBGP (MP-iBGP)

## OVERLAY TECHNOLOGIES

### MODERN PROVIDER NETWORK

**Global Segments:** Known and supported by all SR nodes in the domain. Installed in forwarding tables across the network (e.g. "Forward packet to Node1" according to shortest path to Node1")

**Local Segments:** Defined and installed only on originating node, Not forwarded by others, but must be understood network-wide (e.g. "Forward packet to Node1" according to shortest path to Node1")

**Global segments:** Defined in the SR Global Block (SRGB) and should be consistent across all nodes; **local segments** are defined in the SR Local Block (SRLB) and are specific to the local SR node.

**POLICING VS. SHAPING**

- Policing (Inbound mostly):** Drops packets that exceed configured rate limits
- Shaping (Outbound):** Buffers packets to smooth traffic bursts and conform to profile

- Latency Reduction:** Nearby edge servers reduce round-trip time
- Availability:** Failover and redundancy in case of node failure
- Scalability:** Handles traffic spikes via load balancing
- Cost Optimization:** Reduces backend and transit load on origin
- DDoS Protection:** Edge servers absorb attacks → not all traffic on one server
- Global Load Reduction:** Less long-distance traffic across the Internet

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BGP Internal | 200

**EVPN BGP ROUTING TABLE INFOs**

\* = Would not be there if it was L2 VNI BGP Routing Table

**Route Distinguisher:** 172.16.255.101:32777

**Route Type:** 2

```
Path type: internal, path is valid, is best path, no labeled nexthop
Imported to 2 destination(s)
AS-Path: NONE, path sourced internal to AS
172.16.254.101 (metric 81) from 172.16.255.1 (172.16.255.1)
Origin IGP, MED not set, localpref 100, weight 0
Received label 30010 50000
```

- **VXLAN** is a data plane technology which encapsulates Ethernet frames in UDP datagrams to tunnel layer 2 frames over a layer 3 network.
- The underlay network is unaware of VXLAN devices that connect to the

- physical switches are unaware of VXLAN.
- A route distinguisher is used to uniquely identify a route in combination with the destination prefix.

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